

Module 1: Introduction & System Structures

- **Operating System Services:** Explain the essential services provided by an OS for users and programs.
- **Dual-Mode Operation:** Differentiate between user mode and kernel mode. Explain its importance in system protection.
- **System Calls:** Define a system call. Explain the parameter-passing methods used by system calls.
- **OS Structures:** Discuss Monolithic, Layered, Microkernel, and Hybrid operating system architectures.

Module 2: Process Management & CPU Scheduling

- **Process States:** Describe the lifecycle of a process using a process state transition diagram.
- **Process Control Block (PCB):** Explain the structural components and role of a PCB.
- **CPU Scheduling Algorithms:** Solve numerical problems on FCFS, SJF (Preemptive/Non-preemptive), Priority, and Round Robin scheduling to calculate average waiting time and turnaround time.
- **Inter-Process Communication (IPC):** Compare the shared-memory model and message-passing model.

Module 3: Process Synchronization & Deadlocks

- **Critical Section Problem:** Define the problem and explain the three requirements to solve it (Mutual Exclusion, Progress, Bounded Waiting).
- **Semaphores:** Explain counting and binary semaphores. Outline the classic Dining Philosophers or Producer-Consumer problem using semaphores.
- **Deadlock Characterization:** Detail the four necessary conditions for a deadlock to occur.
- **Banker's Algorithm:** Solve numerical problems for deadlock avoidance to find if a system is in a safe state.

Module 4: Memory Management

- **Paging & Segmentation:** Contrast paging and segmentation architectures with neat hardware diagrams.
- **Page Fault Handling:** Step-by-step explanation of how the OS handles a page fault.
- **Page Replacement Algorithms:** Solve numerical problems using FIFO, LRU, and Optimal page replacement algorithms to find the number of page faults.
- **Thrashing:** Define thrashing, its causes, and how the Working-Set model eliminates it.

Module 5: Storage & File Systems

- **Disk Scheduling Algorithms:** Solve numerical problems to calculate total head movements using FCFS, SSTF, SCAN, C-SCAN, LOOK, and C-LOOK.
- **File Allocation Methods:** Compare Contiguous, Linked, and Indexed file allocation strategies.

- **Directory Structures:** Explain Single-level, Two-level, and Tree-structured directories.
- **Access Matrix:** Explain the mechanism of implementing an access matrix for system protection.

Important topics on module 3, 4, 5

Deadlock detection, recovery, prevention

resource allocation graph, problems, find safe sequences?

page replacement algorithms, problems?

disk scheduling algorithms, problems?

paging, tlb hardware diagram

segmentation

demand paging

page fault handling?

fragmentation

file operations?

access matrix?

directory structures

one level and two level directories

allocation methods

bankers algorithm

resource request algorithm

logical and physical address

copy on write virtual memory